

# GAOPEI PAN

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## RESEARCH/WORK EXPERIENCE

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**University of Würzburg, Germany**

*May.2024 – Present*

PostDoc in Physics

**The University of Hong Kong, China**

*Sept.2023 – Jan.2024*

Research Assistant in Physics (waiting for visa of Germany)

## EDUCATION

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**Institute of Physics, Chinese Academy of Sciences, China**

*Sept.2018 – Jul.2023*

Ph.D. in Physics

**University of Chinese Academy of Sciences, China**

*Sept.2014– Jul.2018*

B.S. in Physics Major with Computer Science Minor

**National University of Singapore, Singapore**

*Jan.2017 – May.2017*

Exchange student in Department of Physics

## RESEARCH SUMMARY

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My research focuses on quantum many-body physics, including non-Fermi-liquid behavior, quantum criticality, correlated phases, and superconductivity in strongly interacting fermion systems and moiré materials. I also work on entanglement-entropy-related problems and am currently exploring gauge-field-coupled Kondo systems. In addition to model studies, I specialize in numerical method development, particularly Quantum Monte Carlo techniques such as determinantal QMC, momentum-space QMC, entanglement-entropy algorithms, self-learning Monte Carlo, and the sign problem in QMC.

(See research statement for details)

## SELECTED PUBLICATIONS

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1. Quasiparticle Spectroscopy of Chiral Charge Order

Jiangchang Zheng\*, Caiyun Chen\*, **Gaopei Pan\***, Xu Zhang\*, Chen Chen, Yuan Da Liao, Ganesh Pokharel, Andrea Capa Salinas, Yizhou Wei, Hoi Chun Po, Ding Pan, Stephen D. Wilson, Zi Yang Meng, Berthold Jäck

[arXiv:2503.19032](https://arxiv.org/abs/2503.19032)

2. Defining a universal sign to strictly probe a phase transition

Nvsen Ma, Jun-Song Sun, **Gaopei Pan**<sup>†</sup>, Chen Cheng, and Zheng Yan

[Phys. Rev. B 110, 125141 \(2024\)](#)

3. Stable computation of entanglement entropy for two-dimensional interacting fermion systems

**Gaopei Pan**, Yuan Da Liao, Weilun Jiang, Jonathan D’Emidio, Yang Qi, Zi Yang Meng

[Phys. Rev. B 108, L081123\(2023\)](#)

4. Thermodynamic characteristic for correlated flat-band system with quantum anomalous Hall ground state

**Gaopei Pan**, Xu Zhang, Hongyu Lu, Heqiu Li, Bin-Bin Chen, Kai Sun, Zi Yang Meng

[Phys. Rev. Lett. 130, 016401 \(2023\)](#)

## PUBLICATIONS

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### First Author/Corresponding Author:

1. Quasiparticle Spectroscopy of Chiral Charge Order  
Jiangchang Zheng\*, Caiyun Chen\*, **Gaopei Pan\***, Xu Zhang\*, Chen Chen, Yuan Da Liao, Ganesh Pokharel, Andrea Capa Salinas, Yizhou Wei, Hoi Chun Po, Ding Pan, Stephen D. Wilson, Zi Yang Meng, Berthold Jäck  
[arXiv:2503.19032](#)
2. High-efficiency quantum Monte Carlo algorithm for extracting entanglement entropy in interacting fermion systems  
Weilun Jiang, **Gaopei Pan**<sup>†</sup>, Zhe Wang, Bin-Bin Mao, Heng Shen, Zheng Yan  
[arXiv:2409.20009](#)
3. Defining a universal sign to strictly probe a phase transition  
Nvsen Ma, Jun-Song Sun, **Gaopei Pan**<sup>†</sup>, Chen Cheng, and Zheng Yan  
[Phys. Rev. B 110, 125141 \(2024\)](#)  
[arXiv:2301.12438](#)
4. Stable computation of entanglement entropy for two-dimensional interacting fermion systems  
**Gaopei Pan**, Yuan Da Liao, Weilun Jiang, Jonathan D’Emidio, Yang Qi, Zi Yang Meng  
[Phys. Rev. B 108, L081123\(2023\)](#)  
[arXiv:2303.14326](#)
5. Thermodynamic characteristic for correlated flat-band system with quantum anomalous Hall ground state  
**Gaopei Pan**, Xu Zhang, Hongyu Lu, Heqiu Li, Bin-Bin Chen, Kai Sun, Zi Yang Meng  
[Phys. Rev. Lett. 130, 016401 \(2023\)](#)  
[arXiv:2207.07133](#)
6. Dynamical properties of collective excitations in twisted bilayer Graphene  
**Gaopei Pan**, Xu Zhang, Heqiu Li, Kai Sun, Zi Yang Meng  
[Phys. Rev. B 105, L121110 \(2022\)](#)  
[arXiv:2102.10755](#)
7. Momentum space quantum Monte Carlo on twisted bilayer Graphene  
Xu Zhang\*, **Gaopei Pan\***, Yi Zhang, Jian Kang, Zi Yang Meng  
[Chin. Phys. Lett. 38, 077305 \(2021\) Cover story](#)  
[arXiv:2105.07010](#)
8. Yukawa-SYK model and Self-tuned Quantum Criticality  
**Gaopei Pan**, Wei Wang, Andrew Davis, Yuxuan Wang, Zi Yang Meng  
[Phys. Rev. Research 3, 013250 \(2021\)](#)  
[arXiv:2001.06586](#)
9. Superconductivity near the  $(2 + 1)$ d ferromagnetic quantum critical point  
Yunchao Hao\*, **Gaopei Pan\***, Kai Sun, Zi Yang Meng, Yang Qi  
[Chin. Phys. Lett. 39 097102 \(2022\)](#)
10. Self-learning Monte Carlo Method: A Review  
**Gaopei Pan**, Chuang Chen, Zi Yang Meng  
[arXiv:2507.12554 \(Book chapter\)](#)
11. A Sport and a Pastime: Model Design and Computation in Quantum Many-Body Systems  
**Gaopei Pan**, Weilun Jiang, Zi Yang Meng  
[Chin. Phys. B 31 127101 \(2022\)](#)

12. Sign Problem in Quantum Monte Carlo Simulation

**Gaopei Pan**, Zi Yang Meng

[arXiv:2204.08777](#) ([Book chapter for Elsevier Encyclopedia of Condensed Matter Physics](#)).

**Non First Author:**

1. Loop-Current Fluctuations Mediated Chiral d-Wave Pairing in Kagome Lattices

Qi-Fang Li, **Gaopei Pan**, Xu Zhang, Satoru Nakatsuji, Weilun Jiang, Xiao Yan Xu and Xianxin Wu

[Chin. Phys. Lett. 42 057302 \(2025\)](#)

2. Reweight-annealing method for calculating the value of partition function via quantum Monte Carlo

Yi-Ming Ding, Jun-Song Sun, Nvsn Ma, **Gaopei Pan**, Chen Cheng, Zheng Yan

[Phys. Rev. B 110, L121112 \(2024\)](#)

[arXiv:2403.08642](#)

3. Universal collective Larmor-Silin mode emerging in magnetized correlated Dirac fermions

Chuang Chen, Yuan Da Liao, Chengkang Zhou, **Gaopei Pan**, Zi Yang Meng, Yang Qi

[Phys. Rev. B 110, L121112 \(2024\)](#)

[arXiv:2401.14358](#)

4. Integral algorithm of exponential observables for interacting fermions in quantum Monte Carlo simulations

Xu Zhang, **Gaopei Pan**, Bin-Bin Chen, Kai Sun, Zi Yang Meng

[Phys. Rev. B 110, L121112 \(2024\)](#)

[arXiv:2311.03448](#)

5. Disorder Operator and Rényi Entanglement Entropy of Symmetric Mass Generation

Zi Hong Liu, Yuan Da Liao, **Gaopei Pan**, Weilun Jiang, Chao-Ming Jian, Yi-Zhuang You, FakherF Assaad, Zi Yang Meng, Cenke Xu

[Phys. Rev. Lett. 132, 156503 \(2024\)](#)

[arXiv:2308.07380](#)

6. Evolution from quantum anomalous Hall insulator to heavy-fermion semimetal in twisted bilayer graphene

Cheng Huang, Xu Zhang, **Gaopei Pan**, Heqiu Li, Kai Sun, Xi Dai, Zi Yang Meng

[Phys. Rev. B 109, 125404 \(2024\)](#)

[arXiv:2304.14064](#)

7. The teaching from entanglement: 2D deconfined quantum critical points are not conformal

Yuan Da Liao, **Gaopei Pan**, Weilun Jiang, Yang Qi, Zi Yang Meng

[arXiv:2302.11742](#)

8. Quantum Monte Carlo sign bounds, topological Mott insulator and thermodynamic transitions in twisted bilayer graphene model

Xu Zhang, **Gaopei Pan**, Bin-Bin Chen, Heqiu Li, Kai Sun, Zi Yang Meng

[Phys. Rev. B 107, L241105\(2023\)](#)

[arXiv:2210.11733](#)

98. Fermion sign bounds theory in quantum Monte Carlo simulation

Xu Zhang, **Gaopei Pan**, Xiao Yan Xu, Zi Yang Meng

[Phys. Rev. B 106, 035121 \(2022\)](#)

[arXiv:2112.06139](#)

10. Superconductivity and bosonic fluid emerging from Moiré flat bands

Xu Zhang, Kai Sun, Heqiu Li, **Gaopei Pan**, Zi Yang Meng

[Phys. Rev. B 106, 184517 \(2022\) Editors' Suggestion](#)  
[arXiv:2111.10018](#)

11. Phase diagram of the spin-1/2 Yukawa-SYK model: Non-Fermi liquid, insulator, and superconductor

Wei Wang, Andrew Davis, **Gaopei Pan**, Yuxuan Wang, Zi Yang Meng  
[Phys. Rev. B 103, 195108 \(2021\)](#)  
[arXiv:2102.10755](#)

12. Solving quantum rotor model with different Monte Carlo techniques

Weilun Jiang, **Gaopei Pan**, Yuzhi Liu, Zi Yang Meng  
[Chin. Phys. B 31, 040504 \(2022\)](#)  
[arXiv:1912.08229](#)

13. Revealing Fermionic Quantum Criticality from New Monte Carlo Techniques

Xiao Yan Xu, Zi Hong Liu, **Gaopei Pan**, Yang Qi, Kai Sun, Zi Yang Meng  
TOPICAL REVIEW, [J. Phys.: Condens. Matter 31, 463001 \(2019\)](#)  
[arXiv:1904.07355](#)

14. Itinerant Quantum Critical Point with Fermion Pockets and Hot Spots

Zi Hong Liu, **Gaopei Pan**, Xiao Yan Xu, Kai Sun, Zi Yang Meng  
[PNAS August 20, 2019 116 \(34\) 16760-16767](#)  
[arXiv:1808.08878](#)